

Air Separation Process



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Senior Design, Mini Project 1

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Project Charter

Title	Goal	In Scope	Out of Scope
Air Separation Process	To choose the optimal design for the synthesis of oxygen with maximum purity	Design of the cryogenic air separation using liquified natural gas (LNG)	• Transportation of Oxygen tanks • Location of plant • Plant safety • Natural gas pretreatment
Deliverables		Team Members	
• Design Plan • Market Analysis • Possible Solutions • Technology comparisons • BFD & PFD		☐ Velia Palma ☐ Ahmed Alghamdi ☐ Erica Caruso ☐ Huda Usman	

Design Plan

Oxygen Processing					
Task ID	Task	Start Date	Duration(Days)	Finish Date	Assignee
Logistics					
1	First Meeting(Task Delegation)	9-Sep	1	9-Sep	All
2	Market Analysis Research	11-Sep	3	14-Sep	Erica Caruso
3	Technology Comparison Research	11-Sep			All
4	BFD & PFD Research	11-Sep	7	17-Sep	All
5	Project Charter	11-Sep	1	11-Sep	Huda Usman
6	Design Plan	9-Sep	1	9-Sep	Huda Usman
15	Layout/Format	11-Sep	7	17-Sep	All
Executive					
Introduction					
7	Objective	11-Sep	7	17-Sep	Huda Usman
8	Market Analysis	11-Sep	7	17-Sep	Erica Caruso
Process Description					
9	Problem	15-Sep	7	17-Sep	All
10	Possible Solutions	11-Sep	7	17-Sep	All
11	Chosen Solutions	11-Sep	7	17-Sep	Velia
12	Design Basis and Assumptions	11-Sep	7	17-Sep	Huda Usman
13	BFD	11-Sep	7	17-Sep	Ahmed
14	PFD & Mass balance	11-Sep	7	17-Sep	Huda Usman
Wrap-Up					
16	Section Unification	16-Sep	1	17-Sep	All
17	Proofreading / Plagiarism Check	16-Sep	1	17-Sep	All

Introduction

Objective

In this report, our focus will be to synthesize oxygen using cryogenic separation technique. This technique is proven to produce oxygen with a purity of 99.5% [4] which then can be distributed to the targeted market. Oxygen is an essential part of our lives. In the early 1800s, oxygen was the only source for patients to support their respiration [3]. For the simplicity of this project, the air around us is assumed to comprise of 21 % oxygen and 79 % Nitrogen [1]. The detailed composition of the air is shown in Table 0 [15].

Chemical	Volume Fraction	Mass Fraction
Nitrogen	78.14	75.6
Oxygen	20.92	23.1
Argon	0.94	0.3
Neon	1.5×10^{-3}	1×10^{-3}
Helium	5×10^{-4}	0.7×10^{-4}
Krypton	1×10^{-4}	3×10^{-4}
Hydrogen	5×10^{-5}	0.35×10^{-5}
Xenon	1×10^{-5}	4×10^{-5}

Table 0. Composition of air

Variety of air separation techniques are used for oxygen applications not only for the use in hospitals and industries but also in academic research [2]. Figure 1 illustrates that due to a dramatic increase in respiratory diseases, the demand for oxygen therapy is expected to increase by 11.1% by 2024 [5].

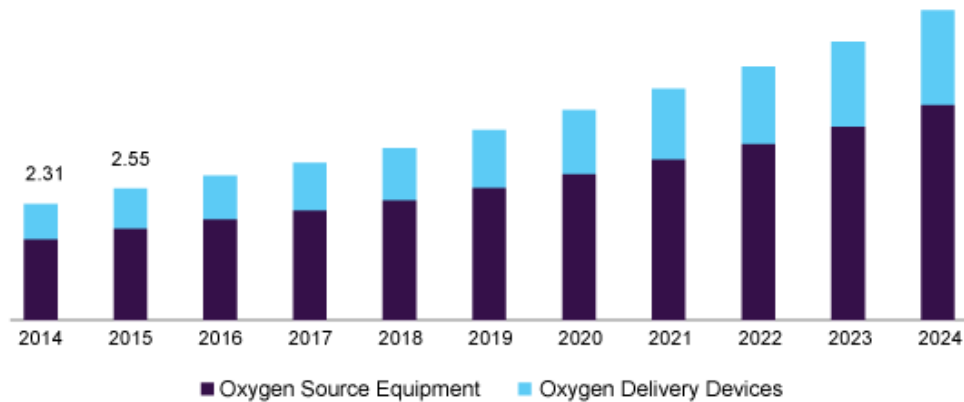


Figure 1. U.S Oxygen Therapy market size increase from 2014-2024 (Grand View Research)

According to a survey conducted by the British Medical Research Council, it was clear that with the increase in respiratory diseases the demand of global oxygen therapy would increase [5]. North America was observed to have higher demand in 2015 and it cost 2,844 US million dollars [5]. Figure 2 illustrates these results below.

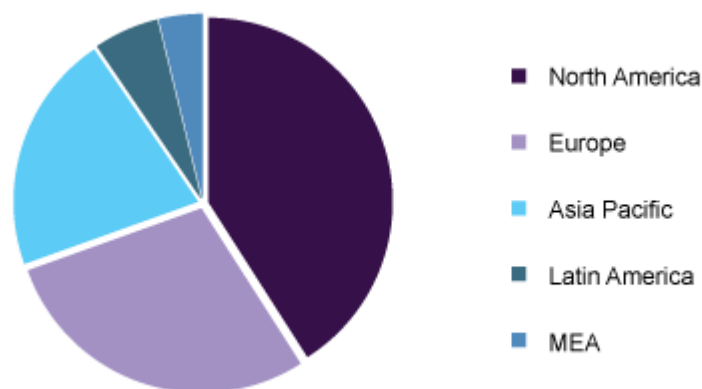


Figure 2. U.S Global Oxygen Therapy Market as of 2015 (Grand View Research)

Therefore, our goal for this project is to efficiently produce oxygen with maximum purity in order to meet the needs of our growing community.

Market Analysis

As mentioned earlier, the growth for oxygen therapy has increased and is predicted to grow dramatically over the next 5 years. The cause for this major growth is due to increased respiratory disease. The demand for home healthcare has also caused this growth in oxygen therapy to occur. In the U.S, approximately 12 million adults have been diagnosed with Chronic Obstructive Pulmonary Disease (COPD) and out of this 12 million people about 120,000 die each year. Asthma is also a major concern today because about 344 million people deal with asthma [5]. Since all these patients rely on oxygen therapy, these high numbers create a huge demand for these products. This is predicted to drive this industry in years to come. According to Zion Market Research, the oxygen therapy market was valued at about 9.4 million dollars. This value is predicted to increase by another 10 million dollars by the year of 2025 [6].

A science direct article studies the economics of specifically cryogenic air separation using NG, which is what we are focusing on for the project. This economic evaluation is based on the concept of Annualized Cost of System (ACS) [8]. The economic analysis was calculated using the equations in Table 1 below.

Parameter	Equation	Ref
Annualized Capital Cost	$C_{acap} = C_{cap}$ $\cdot CRF(i, Y_{proj})$ $= C_{acap} \cdot \frac{i(1+i)^{Y_{proj}}}{(1+i)^{Y_{proj}} - 1}$	[16]
Annual Real Interest Rate	$i = \frac{j-f}{1+f}$	[16]
Replacement Cost	C_{rep} $= C_{cap} \text{ (In Base Year)}$ $\cdot (1+i)^{Y_{rep}}$	[16]
Annualized Replacement Cost	$C_{arep} = C_{rep}$ $\cdot SFF(i, Y_{rep})$ $= C_{rep} \cdot \frac{j}{(1+i)^{Y_{rep}} - 1}$	[16]
Net present value (NPV)	NPV $= \frac{ACS}{CRF(i, Y_{proj})}$ $= ACS \cdot \frac{(1+i)^{Y_{proj}} - 1}{i(1+i)^{Y_{proj}}}$	[16]
Levelized cost of product (LCOP)	$LCOP$ $= \frac{ACS}{\text{Annual output product of the system}}$	[16]

Table 1. Economic parameters which is economic analysis

Using these equations, the economic analysis was able to be performed and the results were calculated. Some assumptions were made to make these calculations including market prices for oxygen and nitrogen. The assumed prices of oxygen and nitrogen were \$0.05 and \$0.02 per kg. The study also assumed that the electrical energy price was \$0.15 per kWh and the labor cost was 400 dollars per month [8]. The study shows the effect of using an air separation unit (ASU) with and without liquified natural gas (LNG). Table 2 below shows the results from this economic analysis and the improvement with using LNG.

Parameter (ASU)	Value	Parameter (Integration of ASU with LNG)	Value	Percentage improvement
Capital Cost (US\$)	111,370,132	Capital Cost (US\$)	92,378,895	17.05
Electrical Energy Cost (US\$)	30,383,214	Electrical Energy Cost (US\$)	27,940,370	8.04
Net Annual Benefit (US\$)	28,266,531	Net Annual Benefit (US\$)	31,528,633	11.54
Net Present Value (US\$)	562,969,594	Net Present Value (US\$)	500,749,270	11.05
Annualized cost of system (US \$)	2278293	Annualized cost of system (US \$)	-2714720	219.28
Levelized cost of product (US \$ per Kg oxygen)	0.00446	Levelized cost of product (US \$ per Kg oxygen)	-0.00532	219.28
Operating and Flow Cost (US\$ Per Year) (Without N2 Benefit)	37,495,009	Operating and Flow Cost (US\$ Per Year) (Without N2 Benefit)	33,858,578	9.70
Operating and Flow Cost (US\$ Per Year) (With N2 Benefit)	-5,910,440	Operating and Flow Cost (US\$ Per Year) (With N2 Benefit)	-9,566,494	61.86
Period of Return (Year)	3.94	Period of Return (Year)	2.93	25.63
Rate of Return (Percent)	25.38	Rate of Return (Percent)	34.13	34.48

Table 2. Comparing results of ASU economic analysis before and after the integration of LNG



It is important to compare the gas prices and to see the improvement with using the integration of LNG. Figure 3 shows the effect of oxygen price on the period of return using the nitrogen price to be \$0.01 per kg. This graph shows that ASU with an integration of LNG creates a reduction of 12.43 years in the period of return when the price of oxygen is 4 cents per kg, and it also shows a reduction of 0.87 years when oxygen is priced at 10 cents per kg. These results demonstrate how dramatic this integration affects the market price of oxygen.

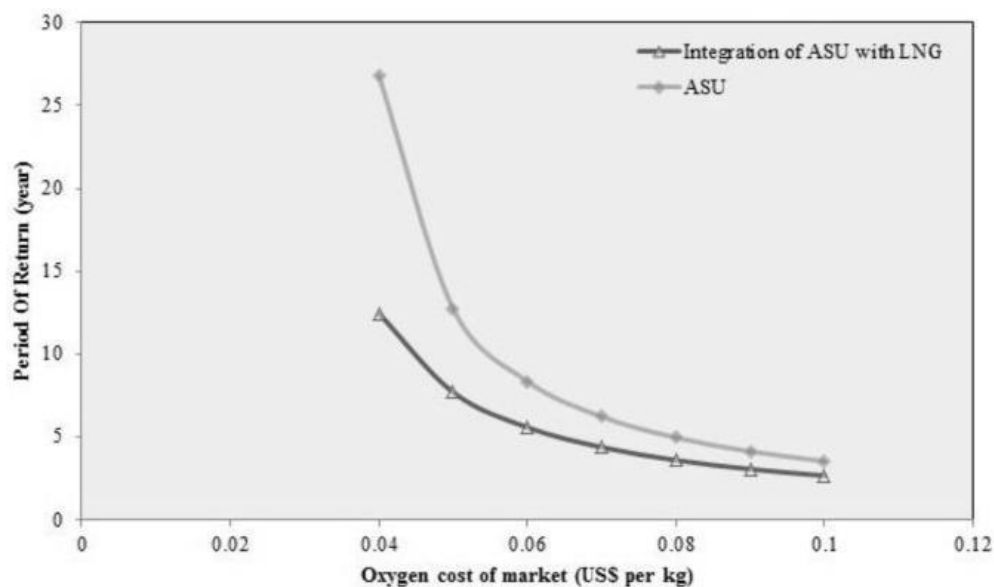


Figure 3. The impact of oxygen on the period of return before and after the integration of LNG in stream in price of 0.01 per kg nitrogen

Process Description

Problem

Producing oxygen with maximum purity is of great importance in air separation processes. Dust particles or other small substances can drastically affect the nature of the process in which oxygen is used. According to the literature reviews our team conducted, the following methods are used to separate air and produce oxygen to achieve maximum purity: cryogenic air separation, membrane technology, adsorption and vacuum technology.

Air separation using **cryogenic separation** is widely used in industries because of its cost-effectiveness and feasibility. Cryogenic separation of air is the process used to produce higher quantities of oxygen, nitrogen, and argon. In this process, pre-treated air is first compressed at ambient pressure of around 6 bar. Next, filtration and the after-cooling process remove any contaminants such as hydrocarbons, water, and carbon dioxide. The carbon dioxide and water are removed in order to

avoid the formation of dry ice or ice during the cooling procedure. Next, the air is cooled to about -173°C [10] and distilled to collect the oxygen, nitrogen and sometimes argon if needed. Refrigeration is conserved by using the heat exchangers to warm the product streams against the air feed [7]. Depending on the desired product qualities and the desired number of output streams, the number of distillation columns and heat exchangers can be varied to separate air. However, by selecting these alternatives, the power consumption and the cost of running the plant could also increase. The conventional cryogenic process diagram is shown in Figure 04 below.

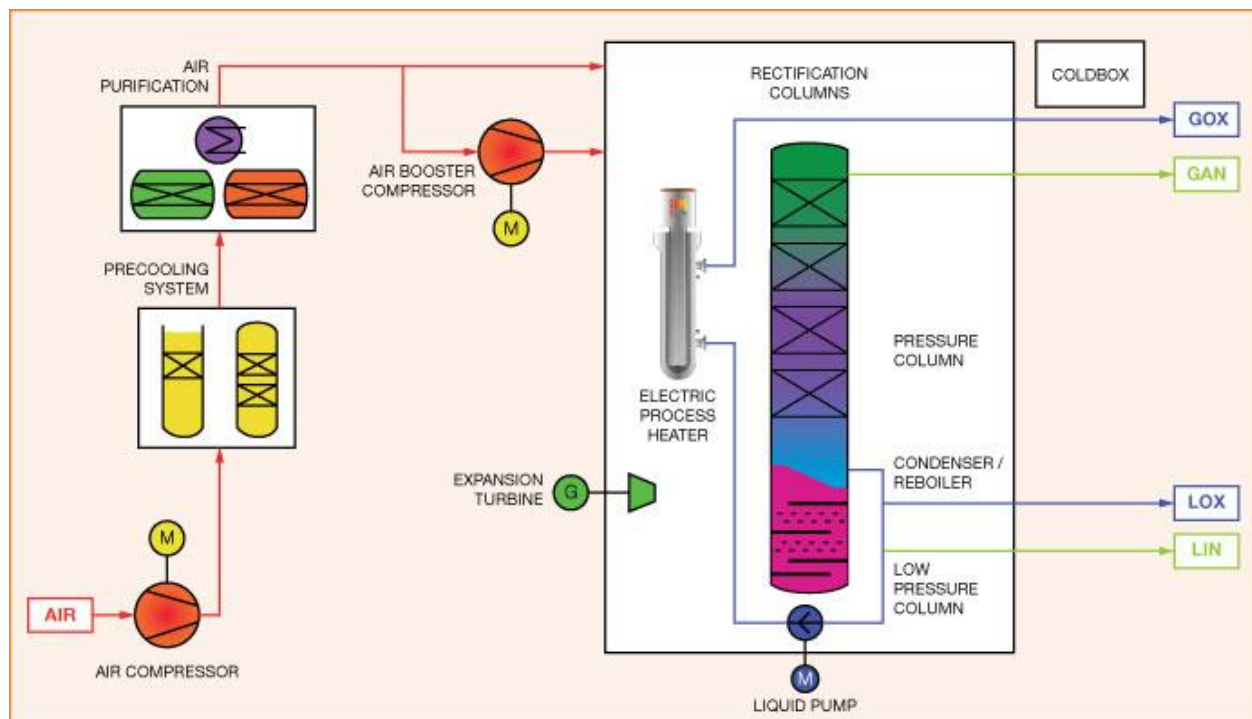


Figure 4. Process flow diagram of conventional cryogenic process.

Membrane Technology is an alternative way of producing Oxygen. Unlike other processes, which require high capital investments, membrane technology does not require as much energy as the others. Membrane technology is typically used to produce industrial nitrogen

and only small quantities of oxygen. The process is designed to be fully operating under the absence of labor. It is a simple process, where every step is being done automatically using a high-tech unit gas separation module, called membrane cartridge, composed of hollow-fiber membranes. Whereas, the membrane acts as a sharp barrier allowing desired components to pass through while recalling the others. “As cleaned compressed air passes through the modules and travels along the entire length of each fiber, it covers a tremendous surface area and is separated into nitrogen and oxygen by the selective membranes” [13]. The membrane plant also consists of other important equipment, such as vacuum pump, pre-purifier unit, and the plant control system.

Oxygen can be also separated using **Pressure Swing Adsorption** technique. This process is well known in chemical industries as pressure swing adsorption (PSA), vacuum swing adsorption (VSA) systems, and hybrid vacuum-pressure swing adsorption (VPSA) systems. The purpose of the process is to separate air into desired products. In this project, oxygen is the desired product. In a PSA process the atmospheric air is compressed up to the required pressure in a large compressor. The air then passes through filter package to hold off dust particles before it goes to coal tower for more filtration. The processes “have relied almost entirely on activated-carbon-based molecular sieves (also called sorbents) to capture oxygen and produce nitrogen with 95 to 99.5 % purity, or alumina in combination with zeolite silicates to trap nitrogen in order to produce oxygen with purities from 90 to 95 %” [13].

After that, oxygen and nitrogen get separated using zeolites beds in unit called PSA generator. Under high pressure, zeolites beds tend to attract nitrogen molecules from compressed air, allowing oxygen to be separated. At that point, raw oxygen is repressured using

booster pump before it goes to the oxygen receiver for purification purposes. Another PSA unit filled with zeolites beds would give high purity for desired product up to 99.5%. However, PSA does not have large scale of production. Thanks to improvements in absorbers and decreases in energy requirement the PSA systems of today can regularly produce around 200 tons/d of oxygen.

Possible Solutions

There is little market demand for oxygen of less than 95% purity therefore the cryogenic process presents the most viable solution to the production of high purity and tonnage of oxygen gas. Table 3 compares the oxygen production methods discussed above as well as others and demonstrates the advantages of the Cryogenic method in economic range, byproduct capability and purity limit.

Process	Status	Economic range (sTPD)	Byproduct capability	Purity limit (vol.%)	Start-up time
Cryogenic	Mature	>20	Excellent	99+	Hours
Adsorption	Semi- mature	<150	Poor	95	Minutes
Polymeric membrane	Semi- mature	<20	Poor	~40	Minutes
Chemical	Developing	Undetermined	Poor	99+	Hours
Ceramic membrane	Developing	Undetermined	Poor	99+	Hours

Table 3. The comparisons of process alternatives for oxygen production from air separation [14]

While cryogenic distillation technology is noted to be complex, expensive and energy intensive; there are a variety of modifications that can make the process efficient and should be taken into consideration while designing a plant. Some of these alternatives are cryogenic air separation with flash separator and cryogenic air separation with Liquefied Natural Gas (LNG) as a heat sink.

Cryogenic separation with flash separator:

In this separation process, the air is compressed from atmospheric pressure to 4 bar instead of 6 bar before splitting into two streams. The higher stream split is compressed to 6 bar and then cooled in heat exchanger 1 before being introduced into High-Pressure Column. However, the lower stream split is cooled in heat changer 1 and then using a throttle valve its pressure is dropped to 1.3 bar to be introduced into the flash drum, shown in Figure (5) [9]. The oxygen-rich air is collected from the bottom of the drum and the nitrogen is collected from the top as a waste product. Next, the oxygen-rich bottom product is introduced into the Low-Pressure Column.

All other steps in this process are similar to the conventional cryogenic separation. However, in this process, there is no significant difference between the energy consumed by conventional cryogenic technique or this method using a flash separator.

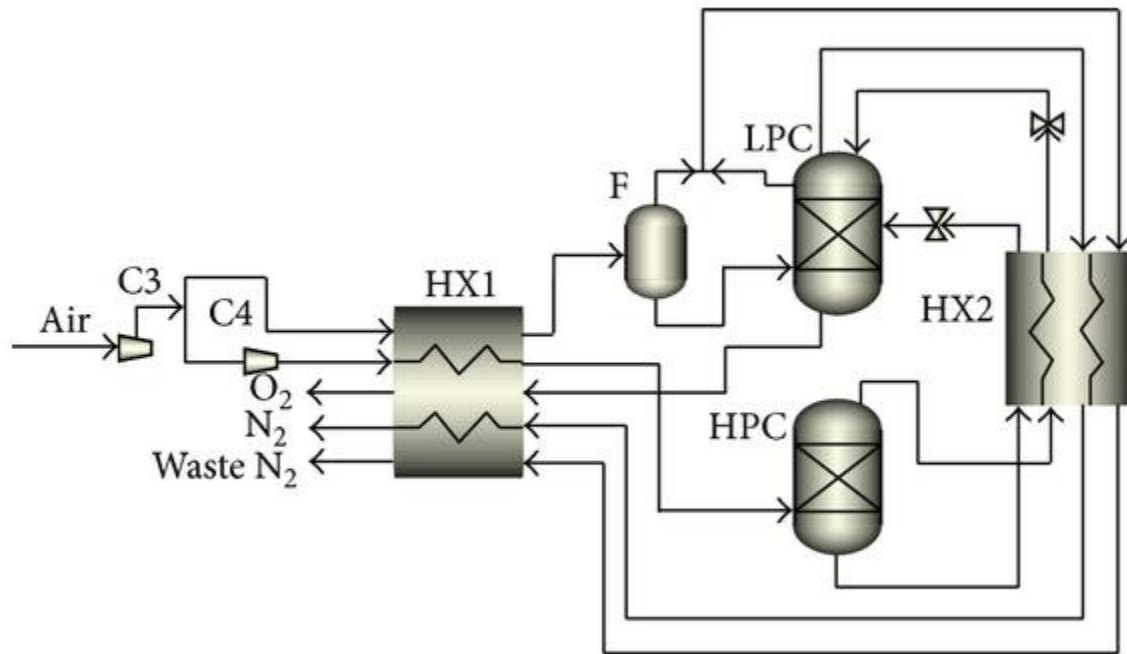


Figure 5. Process flow diagram of cryogenic process with flash separator.

Cryogenic air separation with LNG as a heat sink:

This is a cryogenic air separation process with the integration of liquefied natural gas (LNG). There is an LNG stream added into the process which is clean and efficient energy source. It helps remove any impurities to make the final product as pure as possible. LNG is mostly composed of methane but it is a multicomponent mixture. The natural gas is liquefied at cryogenic liquefaction units and then it is stored at ambient pressure and the temperature is about -162°C [8]. Figure 6 and 7 below shows the differences in the processes with and without LNG. After added LNG into the process, the third compression of the process (Comp 3) is completely removed and also the throttle valves before the bottom distillation column is removed [8]. Because the process is consuming LNG cold energy (around -163°C), the required pressure of the inlet air is decreased. The LNG stream will also change

heat exchanger 1 and will cause it to relax more. The rest of the process is very similar to the process without the integration of LNG.

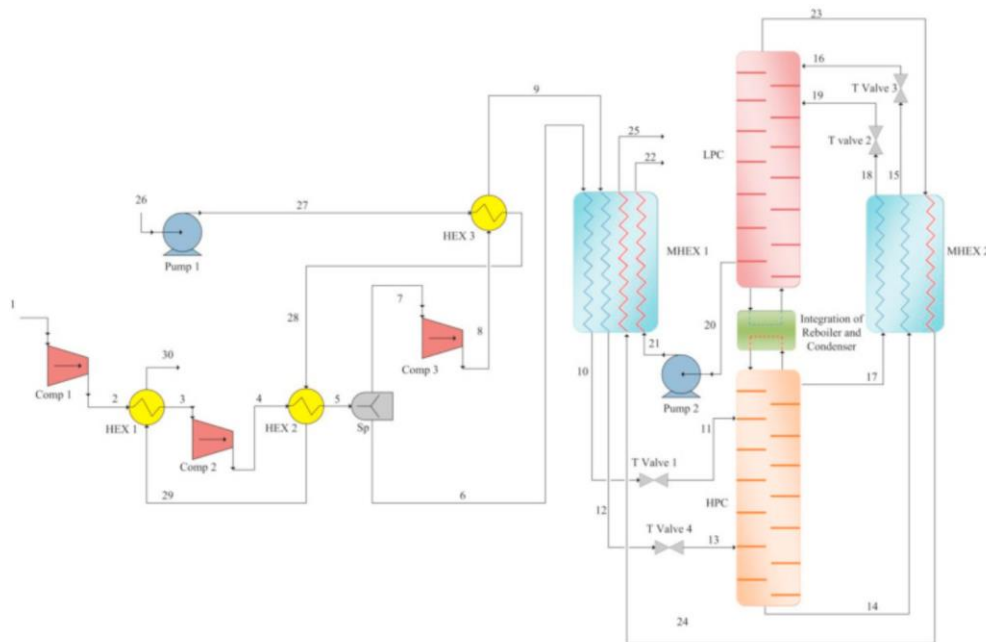


Figure 6. The scheme of two columns cryogenic air separation unit

Chosen Solution

After thorough literature review our group concluded that two-column cryogenic separation using liquified natural gas as shown in Fig. 7 is the preferred solution for air separation and oxygen production. Cryogenic Separation is the industry standard for tonnage quantities and high purity oxygen production. The introduction of LNG to the cryogenic process “will permit a 8.04% reduction in the amount of power requirement and also 17.05% reduction in initial capital cost of ASU plant” [8] when compared to a non LNG integrating plant.

Many other specifications and component changes are being developed and explored “to reduce energy consumption, streamline project execution, and reduce both capital and operating costs, [through the use of] packed towers instead of the traditional trayed towers, moreover others are pursuing heat exchangers, high-efficiency compressors and control systems that provide real-time optimization. The use of higher operating pressures in order to reduce overall column diameter is also being investigated” [13]. However, none of these alternatives have proven more profitable or efficient (for our purposes) than the use of LNG, which is why we have chosen not to implement them at this time.

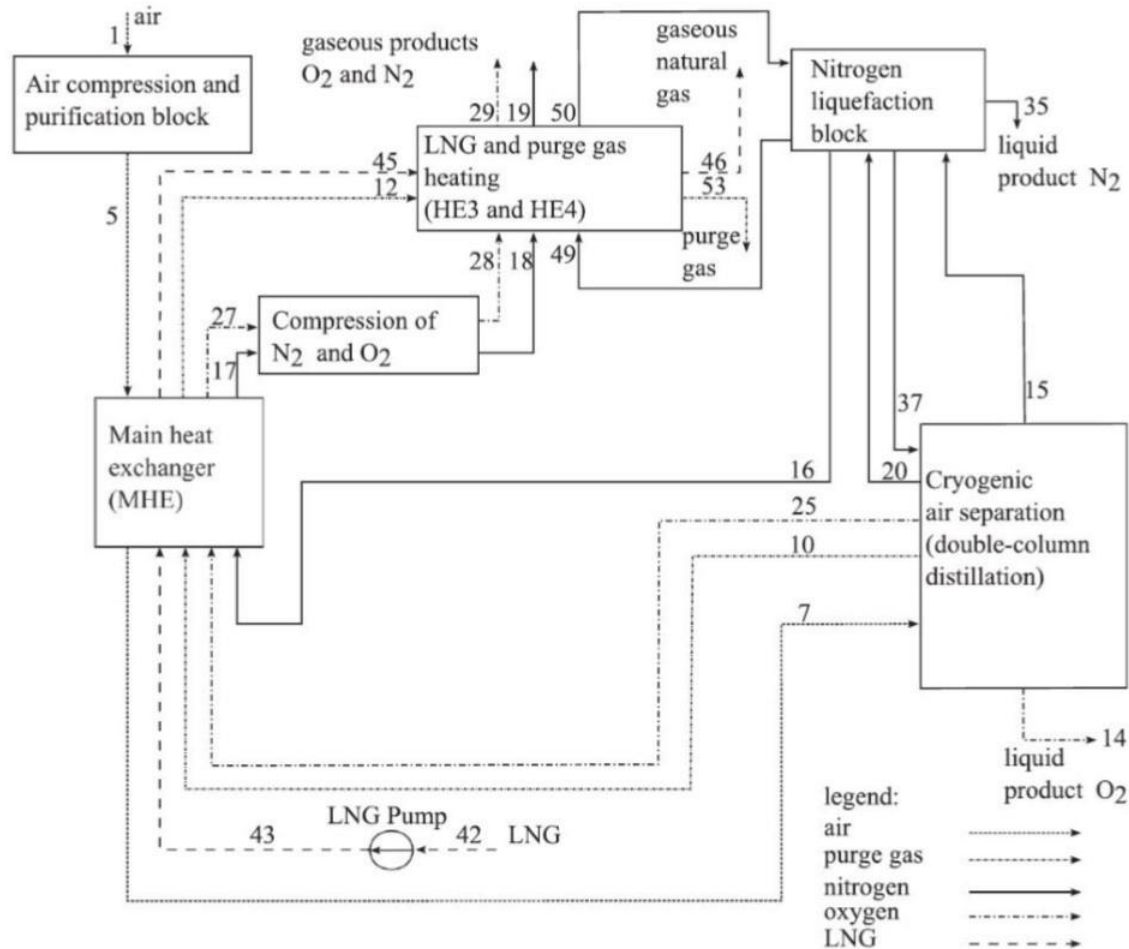


Figure 7. Conceptual design 1 - two-column ASU with LNG integrated in the MHE and gaseous product HE, Tesch et al. (2016)[15]

Here we consider some of the aspects of integration LNC that affect the results of the separation process. The Cryogenic separation process requires a cool down “to around -170 °C to -190 °C, which matches the available cold temperatures of LNG”. Introducing liquid natural gas, which is stored at around 130 kPa and -162 °C[15], into our heat exchanger will recover the cold exergy in the regasification process.

The most important aspect in the integration of the LNG to the process is the matter of deciding in which heat exchanger the exergy is recovered. “In the main heat exchanger (MHE) directly cooling down the air feed” [15], or in the nitrogen loop heat exchangers (HELoop), or in all HEs over the process” [15].

Models have shown that the highest reduction in work requirements is observed when the flow rates through the nitrogen loop are low as it has the highest energy requirement. This can be achieved when integration occurs directly in the MHE “where it cools the cyclic nitrogen stream down to dew point temperature before going to a gas-liquid flash separator” [15].

Flow rate is also an important determinant of efficiency. In many of the reviews studied the flowrate and pressure of the LNG feed varies greatly, from less than 3 up to 113 tons per hour. An intermediate flow rate will be selected in our design basis section and an air flow rate will be determined in accordance.

At the end of this cryogenic separation it is expected to have fulfilled the cold duty requirements of the ASU, to have reduced the work requirements and have produced high quality air. The process results in nitrogen product with high purity (99.99 mol %) and liquid oxygen (99.8 mol % pure) recovered at the bottom of the LP column.

Design Basis

Due to the growth in research and development, there is a huge increase in research about cryogenic processes that are focused on a variety of new applications. Due to the increase in demand from industries, academia, and health care the production and supply of the pure oxygen is crucial [16]. Air Liquide, the world's largest oxygen production company, can produce more than 45000 tons O₂ per day [17]. The design basis of this project aim to produce 10% of this growth capacity. With the assumed 350 days of running plant, our designed process will be able to generate 0.33 lbmol/s oxygen

$$\frac{157,500 \text{ ton } O_2}{\text{year}} \cdot \frac{1 \text{ year}}{350 \text{ days}} \cdot \frac{1 \text{ day}}{24 \text{ hours}} \cdot \frac{1 \text{ h}}{3600 \text{ sec}} \cdot \frac{2000 \text{ lb}}{1 \text{ ton}}$$
$$= 10.42 \text{ lb } O_2 / s = 0.33 \text{ lbmol } O_2 / s$$

Operating Conditions and Assumptions

1. **Design Basis:** 0.33 lbmol/s oxygen produced
2. For simplification purposes, the composition of the air consist of N₂ and O₂ only
3. Generation of Nitrogen is not calculated since not needed during the mass balance.
4. Molecular weight of the air neglecting other gases is taken to be 28.5 lb/mol
5. From [8] the conversion of O₂ is stated to be 100%, however for this project it is taken to be 99.99% conversion.

Block Flow Diagram

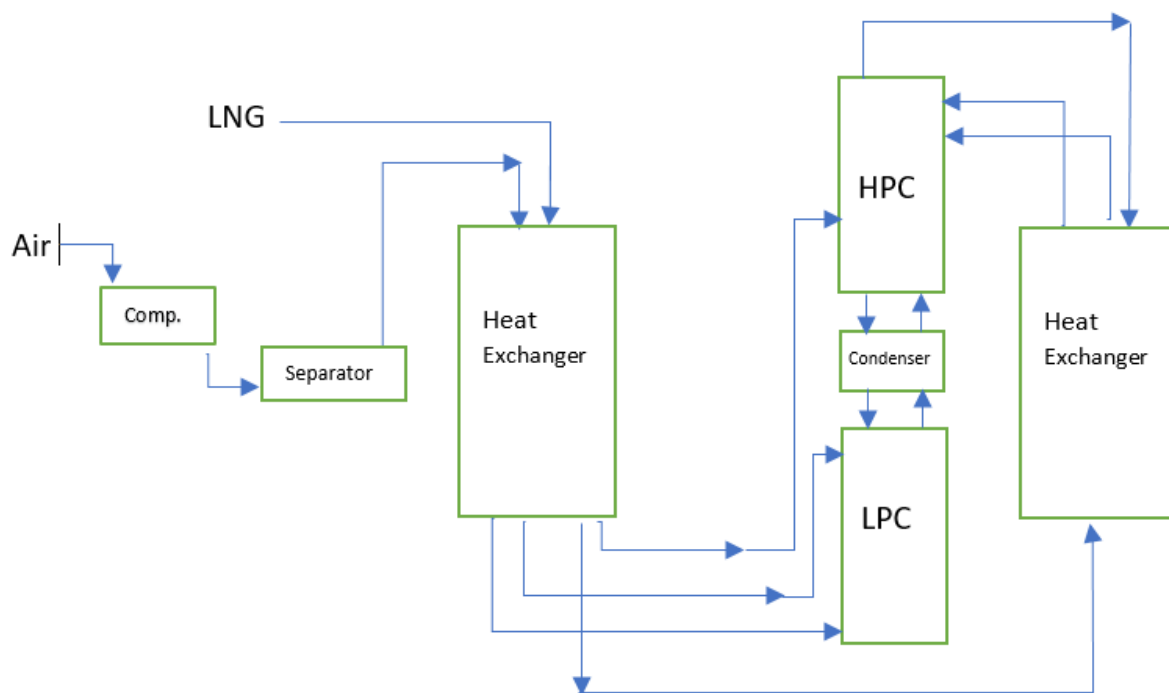


Figure 8. Block Flow Diagram of the cryogenic unit integrated with liquified natural gas

Process Flow Diagram and Mass Balance

Mass Balance:

From our design basis, the production capacity of the plant is 157,500 ton O₂/year with 99.99% conversion.

Therefore the overall mass balance gives,

$$0.9999 \cdot n_{O_2}(IN) = 157,500 \frac{\text{ton } O_2}{\text{year}}$$

$$n_{O_2}(IN) = 157516 \frac{\text{ton } O_2}{\text{year}} = 0.3255 \frac{\text{lbmol } O_2}{s}$$

Since, air consist of 21% oxygen we have,

$$0.21(\text{air}) = 157516 \frac{\text{ton } O_2}{\text{year}}$$

$$m_{\text{air}}(IN) = 750075 \frac{\text{ton air}}{\text{year}} = 1.74 \frac{\text{lbmol}}{s} \text{ of air must be supplied}$$

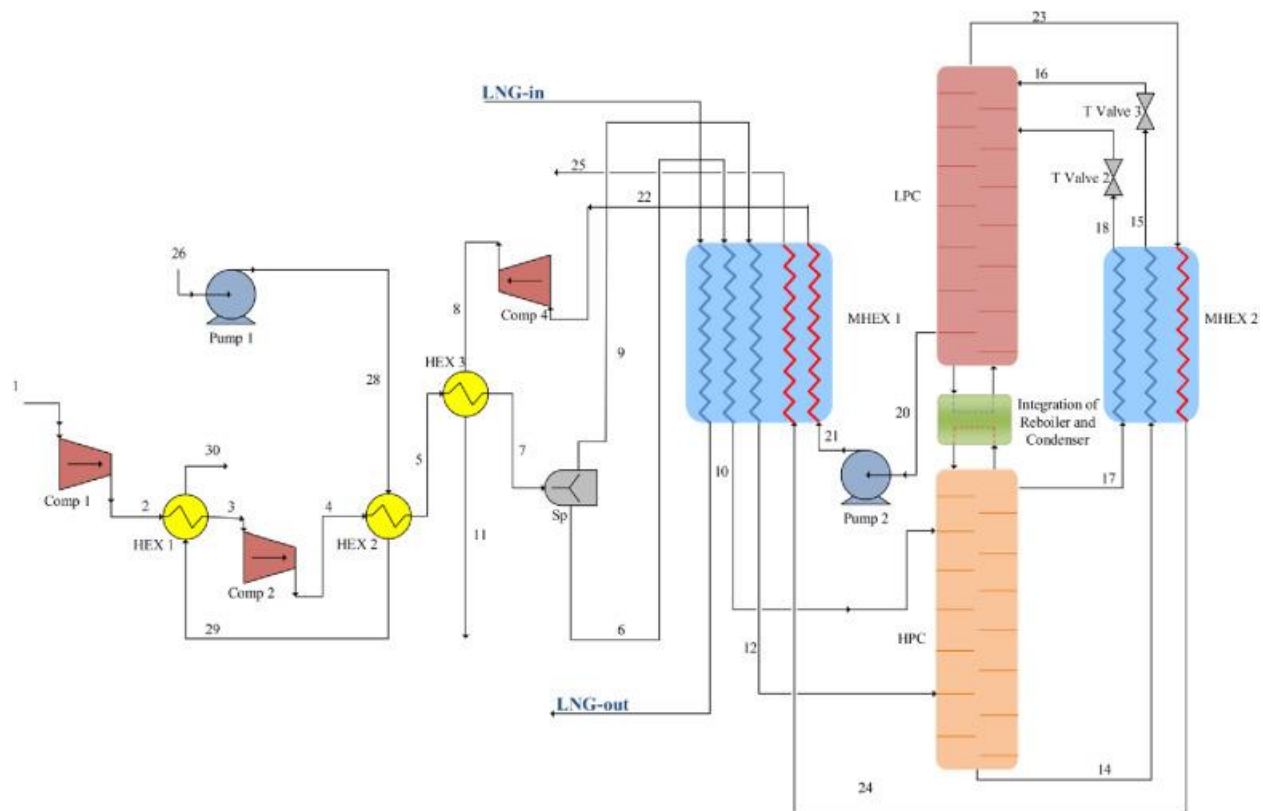


Figure 9. The scheme of two columns cryogenic air separation unit integrated with liquified natural gas

Purposes of The Equipment Used

- 1. Compressors:** a device that increases the pressure of a gas by reducing its volume [11]. The compressors are operated at a pressure ratio around 1.5-2.5 for this LNG process. The compressor pressurizes the gas before and after the heat exchangers to reach the optimal pressure. The pressure of the inlet air increases during the 3 compression stages.
- 2. Heat Exchangers:** device used to transfer heat between two or more fluids. The fluids can be single or two phase and, depending on the exchanger type, may be separated or in direct contact [12]. The heat exchangers for this process are usually operated around 10 kPa. The heat exchanger is used to improve the compressors efficiency and to decrease the work during the compression stages.
- 3. Pump:** a device that moves fluids (liquids or gases), or sometimes slurries, by mechanical action. The pump has a pressure ratio around 2-3 during the LNG process. The pump is necessary to keep the fluid moving through the process.
- 4. Multi component heat exchangers:** heat exchanger with the ability to handle components in multiple phases (solids, liquids, and gases). The multicomponent heat exchanges is operated at the same pressure as the previous heat exchangers and they are used for the same reasons.
- 5. Throttle valve:** a valve designed to regulate the supply of a fluid (as steam or gas and air) to an engine and operated by a handwheel, a lever, or automatically by a governor. The throttle valves are used to regulate the distillation process pressure during this LNG process.
- 6. LPC (Low Pressure Column):** The low pressure column has 60 stages during this process and has a pressure drop of 0 kPa. The reboiler from this column is integrated with the condenser from HPC.
- 7. HPC (High Pressure Column):** The high-pressure column has 50 stages and has a pressure drop of 130 kPa. The cool air enters the column and the first distillation takes place.

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